

Curriculum Vitae

Christopher Marais PhD Student in Computational Biology, University of Florida

Gainesville, Florida, USA

Phone: +1 929-582-0561

Email: gmarais@ufl.edu | cmarais.us@gmail.com

LinkedIn: www.linkedin.com/in/christopher-marais

ORCID: [0000-0003-3409-0473](https://orcid.org/0000-0003-3409-0473)

GitHub: www.github.com/ChristopherMarais

Websites: www.gcmaraais.com

EDUCATION

Doctor of Philosophy (PhD), Computational Biology *University of Florida* (August 2023 – Present)

- **Major:** Forest Resources & Conservation (AI-based applications of computer vision in pest detection for forest health and pest management)
- **Certificates:**
 - Biological Systems Modelling (August 2023 – August 2025)
 - Machine Learning (August 2023 – Present)
- **Relevant Doctoral Coursework:**
 - Neural Net Deep Learning (EEL 6814) – advanced study of deep learning and neural network methods
 - Reproducible Quantitative Methods (FOR 6934) – emphasizes reproducible workflows and advanced data analysis
 - Biological Systems Modeling (ABE 5643C) – computational modeling of biological processes and systems
 - Advanced Research (FOR 7979) – applying machine learning, image analysis, and computational methods to entomological data

Master of Science (MS), Computational Biology *University of Florida* (January 2022 – August 2023)

- **Major:** Forest Resources & Conservation
- **Thesis:** Digital monitoring and automated identification of bark and ambrosia beetles using deep learning

- **Graduated with Highest Honors (GPA 3.95); Awarded August 15, 2023**
- **Relevant Master's Coursework:**
 - Fundamentals of Machine Learning (EEL 5840)
 - Statistical Machine Learning (STA 6703)
 - Image Processing for Remote Sensing (SUR 5386) – remote sensing, hyperspectral imaging methods
 - Topics in Machine Learning (ABE 6933) – in-depth coverage of unsupervised learning and advanced ML techniques
 - Bayesian Statistics (FNR 6560) – Bayesian data analysis for ecological and agricultural datasets

Master of Information Technology (MIT), Big Data Science *University of Pretoria* (2019 – 2020)

- **Grade:** Distinction
- **Thesis:** Interspecies gene network transfer through knowledge graph alignment (incorporating bioinformatics, data analytics, and scientific programming)

Bachelor of Science Honours (BScHons), Bioinformatics *University of Pretoria* (2018)

- **Thesis:** Genome-wide SNP Genotyping for Tropical Pine Tree Species

Bachelor of Science (BSc), Human Biology *University of Pretoria* (2014 – 2017)

- **Majors:** Human Genetics, Psychology, and Human Physiology

PROFESSIONAL CERTIFICATIONS & TRAINING

- **Inclusive Mentoring Academy in Graduate Education (IMAGE)** (University of Florida, Issued Jul 2025)
- **Fundamentals of Deep Learning** (NVIDIA, Issued Jul 2025)
- **Generative AI with Diffusion Models** (NVIDIA, Issued Mar 2024)
- **Getting Started with Deep Learning** (NVIDIA, Issued Feb 2024)
- **Bark & Ambrosia Beetle Academy** (UF/IFAS, Issued Oct 2022)
- **Neo4j Certified Professional** (Neo4j, Issued Oct 2021)
- **Python Programmer Track** (DataCamp, Issued Oct 2021)
- **Neo4j 4.0 Certified** (Neo4j, Issued Sep 2021)
- **Neo4j Graph Data Science Certified** (Neo4j, Issued Sep 2021)
- **Higher Exercises Science and Personal Training** (Enterprises University of Pretoria, Issued Nov 2016)
- **Introduction to Neuroscience Coaching** (Enterprises University of Pretoria, Issued Feb 2016)
- **Neurosciences Visual Skills Testing and Training** (Enterprises University of Pretoria, Issued Feb 2015)
- **CPR Course**

EXPERIENCE

Lead Software Developer *Bark and Ambrosia Beetle Gallery Image Repository* (2025 – Present)

- Developed and maintained the "Bark and Ambrosia Beetle Image Repository," a curated platform for searching, uploading, and downloading specimen data and images of Scolytinae and Platypodinae.
- **Key Responsibilities:** Database creation, data management, and implementing CI/CD pipelines using Docker for cloud computing infrastructure.
- **Project Scope:** Sponsored by the National Science Foundation and USDA APHIS, facilitating global access to curated beetle data.

Journal Peer Reviewer *Journal of Undergraduate Research (JUR), University of Florida* (2025)

- Editorial board member and double-blind peer reviewer for undergraduate research articles in STEM and other disciplines.

Journal Peer Reviewer *Journal of Plant-Environment Interactions* (June 2024 – Present)

- Reviewing manuscripts related to plant-environment interactions and computational applications.

Nucleate Activator *Biotech Accelerator Program* (2025 – Present)

- Participant in a flagship, equity-free biotech accelerator designed to help academic trainees transform scientific research into viable startup companies.
- **Team Formation:** Collaborating with contributors to build a well-rounded founding team.
- **Venture-Building Curriculum:** Participating in workshops on intellectual property (IP), regulatory pathways, clinical needs, and business strategy.
- **Expert Mentorship:** Receiving weekly guidance from industry veterans, venture capitalists, and experienced biotech founders.

Web Developer *Bark Beetle Gallery* (2024)

- Web development for the Bark Beetle Gallery project.

DevOps Engineer / Data Scientist *University of Florida* (Jan 2024 – Present)

- Developing robust data pipelines and CI/CD workflows for Guana River Estuary monitoring.
- Integrating GIS, sensor data, and hydrological modeling data to support decision-making and information sharing tools for water and soil conservation efforts.
- **Skills:** Production Deployment, CI/CD, R, Dashboards, GitFlow.

Graduate Research Assistant *University of Florida* (Jan 2022 – Present)

- Leading deep learning efforts for automated insect identification in the Forest Entomology Lab.
- Collaborating with internal UF teams and external partners (including U.S. governmental agencies such as the USDA) to strengthen pest management.
- Research focuses on applying AI and machine learning to large-scale ecological datasets for both academic and applied outreach.
- **Projects:**
 - Bark beetle species image classification using deep learning.
 - Comparative genomics between bark beetles and fungi.
 - Literature mining for knowledge graph creation and large language model fine-tuning.
- **Skills:** Machine Learning, Computer Vision, Front-End Development, Computational Biology, Graph Databases, NLP, Neural Networks, k-means clustering, Transcriptomics, HPC, Cluster Analysis, Decision Trees, Comparative Genomics.

Collaborating Data Scientist *University of Florida* (May 2023 – Present)

- Machine Learning Operations (MLOps) consultant for the Padilla-Coreano Behavioral Neuroscience Lab.
- Developing reproducible computational and statistical methods for high-throughput behavioral data.
- Creating pipelines for neuroscience data processing and general data management strategies.
- **Skills:** Git, Bash, Python, Scripting, Data Management, Algorithms, Project Management, Experimental Design.

Software Engineer & Data Scientist *PANTHERA* (April 2021 – Dec 2021)

- Built and deployed deep learning models for big cat image recognition.
- Integrated remote sensing data to map animal movement corridors, enhancing conservation extension programs.
- Built software tools for in-the-field offline data analysis on multiple devices.
- **Skills:** AWS, Computer Vision, R Shiny, GitFlow, Software Development.

Associate: Machine Learning and AI *South African SDG Hub* (Jan 2019 – July 2024)

- Built a repository of all research articles produced by universities in South Africa.
- Developed fine-tuned language models on Sustainable Development Goals (SDGs) focused research.
- Created COVID-19 vulnerability mapping via GIS analytics to inform the Office of the President in South Africa.
- Coordinated efforts at the intersection of government needs and data-intensive research, offering insights that guided policy creation.

- **Skills:** SDLC, NLP, Team Management, Python, Machine Learning Algorithms.

Data Analysis Consultant *Healthcare Workers Care Network* (June 2020 – Oct 2020)

- Minor data visualization and consulting on data capture and processing to optimize the COVID-19 psychological support platform for medical professionals.

Machine Learning Contractor *University of Pretoria* (Feb 2020 – July 2020)

- **Project:** Congenital condition diagnostic tool from facial images.
- Developed a machine learning tool aimed at predicting congenital conditions from facial images trained on a very limited set of data.

Technical & Research Assistant *Forest Molecular Genetics (FMG), University of Pretoria* (Jan 2019 – Dec 2020)

- Provided HPC and statistical software support for large-scale genomics research.
- Contributed to bioinformatics pipelines for gene expression profiling and comparative genomics.
- **Skills:** NGS, Transcriptomics, Functional Genomics.

Internships & Early Career

- **Laboratory Intern:** *CTHB - FABI - University of Pretoria* (Jan 2017 – Dec 2017). Skills: Agarose Gel Electrophoresis, PCR Primer Design.
- **Summer Intern:** *University Medical Center Groningen (UMCG), Netherlands* (Dec 2016 – Jan 2017). Skills: Medical Genetics, Single Cell Sequencing.
- **Student Laboratory Assistant:** *University of Johannesburg* (June 2016 – July 2016).

TEACHING & MENTORSHIP

Teaching Assistant & Invited Lecturer *University of Florida* (2024 - Present)

- **Teaching Assistant:** Forests for the Future (FOR 2662) (Mar 2024 – May 2024; Sep 2024 – Dec 2024).
- **Invited Lecturer:** Delivered a Guest lecture on the ethical use of AI in forestry and agricultural research for the *Future of Forest Health Seminar* (FOR 6934) (Apr 2024).
- **Data Carpentries Club:** Teaching assistant.
- **Mentorship:** Mentoring 3 undergraduate and graduate students in research, data analysis and software development.

IMAxUF Int'l Admissions Global Gator Connect Mentor *University of Florida* (March 2024 – September 2024)

- Ambassador and first layer of communication for incoming international students.
- **Skills:** Administrative Assistance.

Capstone Project Lead & Teaching Assistant *University of Pretoria* (Jan 2019 – June

2020)

- **Lead for MIT Big Data Science Capstone:** Acted as client and guide for students (Jan 2020 – June 2020).
- **BScHons Statistics Teaching Assistant:** Teaching assistant for high-intensity statistics modules (Jan 2020 – March 2020).
- **3rd Year Biochemistry Teaching Assistant:** Teaching assistant for bioinformatics lab sessions (2019).

KEY SKILLS

- **Machine Learning & AI:** Deep Learning, Neural Networks, Computer Vision, NLP, Self-Supervised Learning, Generative AI, MLOps, Statistical Machine Learning, k-means clustering, Decision Trees, Random Forests.
- **Bioinformatics & Genomics:** Comparative Genomics, Transcriptomics, Functional Genomics, Single Cell Sequencing, NGS, SNP Genotyping, Genome Assembly, Biological Data Analysis.
- **Programming & Dev:** Python, R, R Shiny, Bash/Linux, SQL, Neo4j/Cypher (Graph Databases), Flutter, JavaScript, Git, GitFlow, SDLC.
- **Tools & Platforms:** AWS, Docker, High Performance Computing (HPC), CI/CD pipelines.
- **Data & Design:** Experimental Design, Project Management, Data Management, Dashboards, Geospatial Data Analysis.
- **Languages:** Afrikaans (Fluent), English (Fluent), German (B2 - Intermediate), Dutch (Beginner/Elementary).

DISSEMINATION OF RESEARCH

Peer-Reviewed Publications

- Hulcr, J., Barnes, I., Barnes, M., Gazis, R., Hammerbacher, A., Johnson, A. J., Lynch, S., Lynn, K., **Marais, G. C.**, Mayers, C. G., Nel, W., Villari, C., Wingfield, B., & Wingfield, M. J. (2025). From forest to fungus: a roadmap to bark beetle mycobiome research. *Phytoparasitica*, 53(45). DOI: 10.1007/s12600-025-01246-x.
- **Marais, C.**, Kuo, E., Hulcr, J., & Dias, R. (2025). IBBI: A Python package for the detection and classification of bark and ambrosia beetles. *Journal of Open Source Software*.
- **Marais, G. C.**, Kajtoch, Ł., Stratton, I. C., Johnson, A. J., & Hulcr, J. (2025). Progress in developing a bark beetle identification tool. *PLOS One*.
- Dong, Y., **Marais, C.**, Wang, B., Lin, W., Chen, Y., Li, Y., Johnson, A. J., & Hulcr, J. (2024). Pre-invasion assessment of potential invasive wood borers on North American tree species in Chinese sentinel gardens. *Entomologia Generalis*, 44(4).
- Jackson, C., Christie, N., Reynolds, S. M., **Marais, G. C.**, Tii-kuzu, Y., Caballero, M., Kampman, T., Visser, E. A., Naidoo, S., Kain, D., & others. (2022). A genome-wide SNP genotyping resource for tropical pine tree species. *Molecular Ecology Resources*, 22(2),

695–710.

- Fourie, W., van der Walt, I., Strydom, H., & **Marais, C.** (2019). South African system tracks SDG research. *Nature*, 573(7773), 196–197.

Preprints & Other Publications

- Stratton, I., **Marais, C.**, Johnson, A., & Hulcr, J. (2024). Bark and Ambrosia Beetle Macro Photography for an AI Training Dataset v1. *Preprint*.
- Jiri Hulcr, Ph.D., Andrew J. Johnson, Ph.D., and **G. Christopher Marais** (2023, September 14). How Systematic Entomology Will Thrive in the Age of Artificial Intelligence. *Entomology Today*.

Theses

- **Marais, C.** (2023). Digital Monitoring and Automated Identification of Bark and Ambrosia Beetles. *Dissertation or Thesis*. University of Florida.
- **Marais, C.** (2020). Interspecies gene network transfer through knowledge graph alignment. *Masters Thesis*. University of Pretoria.
- **Marais, C.** (2018). Genome-wide SNP Genotyping for Tropical Pine Tree Species. *Honours Thesis*. University of Pretoria.

Conference Presentations

- **Marais, C.** (2024, November 12). The Evolution of Classification: AI in the Study of Bark Beetles. Presented at the *Entomological Society of America (ESA) 2024*, Empowering Insect Science, Management and Conservation Through Emerging Technologies Symposium.
- (2020) Development of a High-Throughput Genome-Wide Genotyping Array for Tropical and Subtropical Pine Species. *Plant and Animal Genome XXVIII Conference*, San Diego, CA (January 11–15, 2020).

Posters

- Heeps, T., Cum, M., Illescas-Huerta, E., **Marais, C.**, Iwata, R., Saxena, S., & Padilla-Coreano, N. (2026, March 12). *Social Competition State Encoding in Prefrontal-Hypothalamic Circuits*. Poster presented at Computational and Systems Neuroscience (COSYNE), Cascais, Portugal.
- **Marais, C.** (2023, March 2). Sentinel Garden Data Management. Poster presented at the *Forest Resources and Conservation Graduate Student Symposium*, University of Florida.
- **Marais, C.** (2023, February 15). Sentinel Garden Data Management. Poster presented at the *Emerging Pathogen Institute Research Day*, University of Florida.
- **Marais, C.** (2020). Poster for BScHons research on SNP mining in pine trees, *Department of Bioinformatics Student Research Day*, University of Pretoria.

Media Appearances

- **Marais, C.** & Carney, T. Applications and Efficacy of AI in Forestry. Discussion on the applications of new artificial intelligence in forestry and where it can make an impact.
- **Marais, C.** (Interviewee). (2024). How AI is Changing Insect Identification – Chris Marais. [Interview with Demain Gomez]. *YouTube*.
- Schwartz, B., **Marais, C.**, Stratton, I. (2024). How we Trained an AI to Identify Bark Beetles. *YouTube*.

AWARDS

- **Forest Resource Conservation Outstanding Thesis Award** (University of Florida, Mar 2024)
- **UF IFAS Forest Research Conservation (FRC) Full Funding Graduate Assistantship Award** (University of Florida, Feb 2023)
 - *One of three awards given to extra-ordinary prospective PhD students at the School of Forest, Fisheries, and Geomatics Sciences at the University of Florida for a fully funded 4-year PhD program.*
- **Outstanding International Master's Student** (University of Florida, Sep 2022)
 - *Forest Resources and Conservation Graduate Program's nominee.*
- **Certificate of Outstanding Merit** (University of Florida, MS)
- **Postgraduate Masters Coursework Bursary** (University of Pretoria, Sep 2020)
- **Best BScHons Bioinformatics Research Poster** (University of Pretoria, Jan 2018)

REFERENCES

- Prof. Jiri Hulcr – Associate Professor, Forest Entomology, University of Florida
Email: hulcr@ufl.edu
- Prof. Nancy Padilla-Coreano – Principal Investigator, Padilla-Coreano Lab, University of Florida
Email: npadillacoreano@ufl.edu
- Dr. Geraldine Klarenberg – Lecturer, Quantitative Data Science, University of Florida
Email: gklarenberg@ufl.edu
- Prof. Willem Fourie – Associate Professor, Policy Innovation Lab, Stellenbosch University
Email: willemfourie@gmail.com
- Dr. Ross Pitman – Director, TyzackLabs
Email: rpitman@tyzacklabs.org
- Prof. Zander Myburg – Director, Forest Molecular Genetics, University of Pretoria
Email: zander.myburg@fabi.up.ac.za
- Prof. Fourie Joubert – Associate Professor, Bioinformatics and Genomics, University of Pretoria
Email: fourie.joubert@up.ac.za
- Dr. Vukosi Marivate – Associate Professor, Data Science for Social Impact, University of Pretoria
Email: vukosi.marivate@cs.up.ac.za